

Introduction to Multivariate Analysis

MA2003B Application of Multivariate Methods
in Data Science

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These concepts are fundamental in understanding the relationships between multiple variables in a given dataset.

The Mean Vector and the Covariance Matrix

Definition

Let Π be a population and let

$$X : \Pi \rightarrow \mathbb{R}^p$$

be a random (row) vector. The **mean vector** of X is defined as

$$\mu = \mathbb{E}[X] = [\mathbb{E}[X_1], \mathbb{E}[X_2], \dots, \mathbb{E}[X_p]] = [\mu_1, \mu_2, \dots, \mu_p]$$

where X_i is the i -th component of the random vector X . The **covariance matrix** of X is defined as

$$\Sigma = \text{Cov}[X] = \mathbb{E}[(X - \mu)^T (X - \mu)] = [\text{Cov}[X_i, X_j]]_{i,j=1}^p$$

where $\text{Cov}[X_i, X_j] = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]$ is the covariance between the i -th and j -th components of the random vector X .

Remark

Remember that $\text{Cov}[X_i, X_j] = \text{Cov}[X_j, X_i]$, so the covariance matrix is symmetric. Furthermore, $\text{Cov}[X_i, X_i] = \text{Var}[X_i]$, which is the variance of the i -th component of the random vector X . For this reason, this matrix is also known as the **variance-covariance matrix** of X .

Definition

Given a sample of size n from the population Π , with associated sample matrix x , we define its associated **sample mean vector** as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \left[\frac{1}{n} \sum_{i=1}^n x_{i,1}, \frac{1}{n} \sum_{i=1}^n x_{i,2}, \dots, \frac{1}{n} \sum_{i=1}^n x_{i,p} \right] \quad (1)$$

where x_i is the i -th row of the sample matrix x and $x_{i,j}$ is the j -th component of the i -th row. The **sample covariance matrix** is defined as

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^T (x_i - \bar{x}) \quad (2)$$

where $\hat{\Sigma}$ is a $p \times p$ matrix and $\hat{\Sigma}_{i,j} = \frac{1}{n-1} \sum_{k=1}^n (x_{k,i} - \bar{x}_i)(x_{k,j} - \bar{x}_j)$ is the sample covariance between the i -th and j -th components of the random vector X .

Remark

The sample covariance matrix is an unbiased estimator of the covariance matrix of the random vector X .

Example

Consider the following sample matrix of size $n = 5$ from a population Π :

$$\begin{bmatrix} 2 & 5 & 7 \\ 4 & 3 & 6 \\ 6 & 4 & 5 \\ 8 & 2 & 4 \\ 10 & 1 & 3 \end{bmatrix}$$

Example

To obtain the sample mean vector, we compute:

$$\bar{x}_1 = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

$$\bar{x}_2 = \frac{5 + 3 + 4 + 2 + 1}{5} = 3$$

$$\bar{x}_3 = \frac{7 + 6 + 5 + 4 + 3}{5} = 5$$

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In other words,

$$\bar{\mathbf{x}} = [6, 3, 5]$$

Example

The sample variance of X_j is:

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Therefore,

$$\hat{\Sigma}_{X_1} = \frac{(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2}{4} = 10$$

$$\hat{\Sigma}_{X_2} = \frac{(5-3)^2 + (3-3)^2 + (4-3)^2 + (2-3)^2 + (1-3)^2}{4} = 2.5$$

$$\hat{\Sigma}_{X_3} = \frac{(7-5)^2 + (6-5)^2 + (5-5)^2 + (4-5)^2 + (3-5)^2}{4} = 2.5$$

Example

The sample covariance between X_j and X_k is:

$$\hat{\Sigma}_{X_j, X_k} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$$

This computations can be done in a similar way as before, the results are:

$$\hat{\Sigma}_{X_1, X_2} = -4.5$$

$$\hat{\Sigma}_{X_1, X_3} = -5$$

$$\hat{\Sigma}_{X_2, X_3} = 2.25$$

Example

The resulting covariance matrix is then:

$$\hat{\Sigma} = \begin{bmatrix} 10 & -4.5 & -5 \\ -4.5 & 2.5 & 2.25 \\ -5 & 2.25 & 2.5 \end{bmatrix}$$

The Mahalanobis Distance

Definition

Assume that we have a population Π with a random vector X that has mean vector μ and covariance matrix Σ . If Σ is invertible, then the **Mahalanobis distance** between two points x and y in $\mathbb{R}^p$ is defined as:

$$d_M(x, y) = \sqrt{(x - y)\Sigma^{-1}(x - y)^T} \quad (3)$$

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Given a sample of size n from the population Π , with associated sample matrix x and sample covariance matrix $\hat{\Sigma}$, the **sample Mahalanobis distance** between two points x_i and x_j is defined as:

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Remark

The Mahalanobis distance is a measure of the distance between two points in a multivariate space, taking into account the correlations between the variables. It is particularly useful for identifying outliers in multivariate data.

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We can now construct the vector

$$y = x + \varepsilon$$

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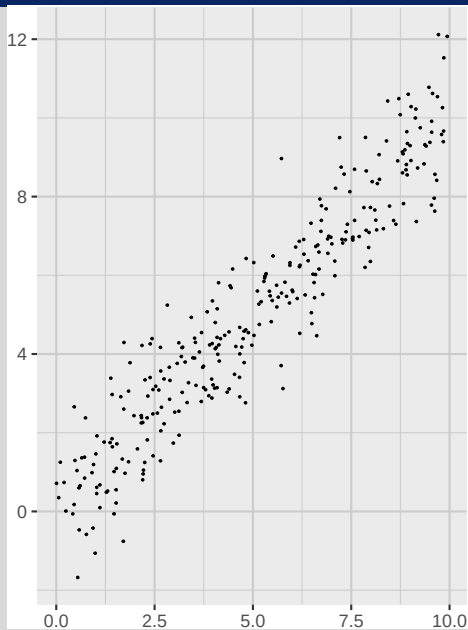
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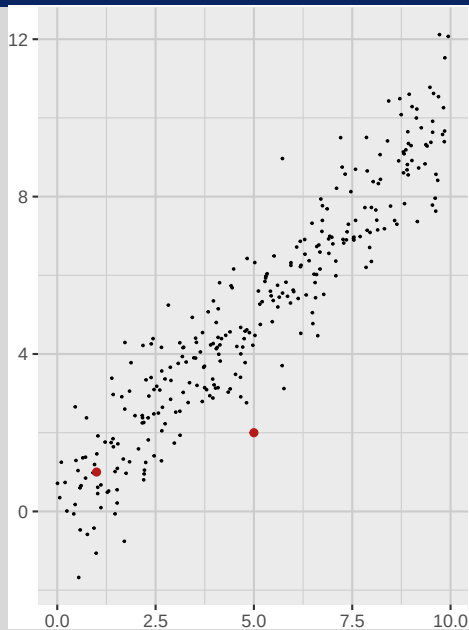
We can now construct the vector

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The associated scatter plot of the data is shown below:



We now add the points $(1, 1)$ and $(5, 2)$ to our data:



Notice that one of these points is clearly an outlier.

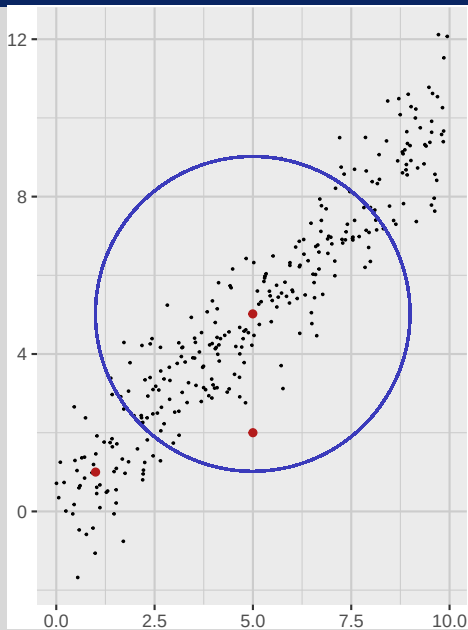
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We would like to have some concrete measure to show this.

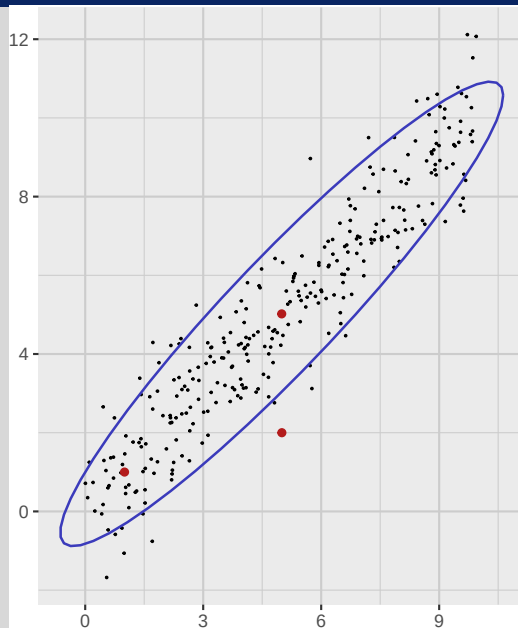
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We would like to have some concrete measure to show this.

However, if we use the Euclidean distance, we obtain that it is actually the outlier point that is closer to the sample mean vector:



On the other hand, if we compute the Mahalanobis distance, we obtain that the outlier point is actually further away from the sample mean vector:



Scaling and Standardization

Now assume that we have a random vector X with mean vector μ and covariance matrix Σ . We can define a new random vector Y as follows:

$$Y^T = AX^T + b$$

where A is a $p \times p$ matrix and b is a p -dimensional vector.

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The mean vector and covariance matrix of the new random vector Y can be computed as follows:

$$\mu_Y^T = \mathbb{E}[Y]^T = A\mu^T + b$$

and

$$\Sigma_Y = \text{Cov}[Y] = A\Sigma A^T$$

As we will show later, it is always possible to find a matrix A and a vector b such that the new random vector Y has mean vector $\mu_Y = 0$ and covariance matrix $\Sigma_Y = I_p$, where I_p is the identity matrix of size $p \times p$. This process is known as a **standardization** of the random vector X .

Activity: The Geometry of Multivariate Data

Part A: Theoretical Exercises

1. Prove that the Mahalanobis distance reduces to the Euclidean distance if the covariance matrix is the identity.
2. Show that Mahalanobis distance is invariant under affine transformations.
3. Show that if you standardize each variable in a multivariate dataset, the covariance matrix becomes the correlation matrix.
4. Assume that X_1 and X_2 are two independent random variables following the standard normal distribution. Show that $d_M(X, 0)^2$ follows a chi-square distribution with 2 degrees of freedom, where $d_M(X, 0)$ is the Mahalanobis distance from the origin.

Part B: Practical Exercises

1. Generate a 2D dataset of 50 points (two correlated variables).
2. Compute the sample mean vector and covariance matrix.
3. Write a function in R that computes the Mahalanobis distance of each point from the mean.
4. Identify the point with the largest Mahalanobis distance.
5. Plot the dataset using `ggplot2`.
6. Draw Mahalanobis distance ellipses for distances 1, 2, and 3.
7. Color points depending on whether their distance is greater than 1, 2, or 3.
8. Create a 3-variable dataset.
9. Compute Mahalanobis distances in 3D.
10. Plot pairs of variables with projected ellipses.

Part C: Advanced Exercises

1. Simulate 5 points in 3 dimensions with one variable as a linear combination of the others.
2. Try to compute Mahalanobis distance and discuss singularity of covariance.
3. Apply a linear transformation to a 2D dataset and verify invariance of Mahalanobis distances.
4. Write an R function that returns points forming a Mahalanobis-distance ellipse.
5. Overlay Mahalanobis-distance ellipses of arbitrary radius on a scatter plot.